

Yoonsung Kim

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Education

KAIST , Daejeon, South Korea <i>Ph.D. Program in Computer Science</i> , Advisor: Jongse Park	Mar. 2023 – Present
KAIST , Daejeon, South Korea <i>M.S. in Computer Science</i> , Advisor: Jongse Park	Feb. 2023
Sungkyunkwan University , Suwon, South Korea <i>B.S. in Computer Science & Engineering</i>	Feb. 2021

Research Interests

- Computer Architecture and Systems
- Low-Precision Quantization Techniques for AI
- Hardware-Software Co-Design in AI Acceleration
- Adaptive AI for On-Device Autonomous Systems

Publications

1. C. Oh, S. Oh, J. Hwang, **Y. Kim**, H. Sharma, J. Park, “Neo: Real-Time On-Device 3D Gaussian Splatting with Reuse-and-Update Sorting Acceleration,” in *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, March 2026. (To Appear)
2. W. Kim, Y. Lee, **Y. Kim**, J. Hwang, S. Oh, J. Jung, A. Huseynov, W. G. Park, C. H. Park, D. Mahajan, J. Park, “Pimba: A Processing-in-Memory Acceleration for Post-Transformer Large Language Model Serving,” in *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, October 2025.
3. J. Hwang, D. Kim, S. Lee, **Y. Kim**, G. Heo, H. Kim, Y. Jeong, T. Meaza, E. Park, J. Ahn, J. Park, “Déjà Vu: Efficient Video-Language Query Engine with Learning-based Inter-Frame Computation Reuse,” in *International Conference on Very Large Data Bases (VLDB)*, September 2025.
4. **Y. Kim**, C. Oh, J. Hwang, W. Kim, S. Oh, Y. Lee, H. Sharma, A. Yazdanbakhsh, J. Park, “DaCapo: Accelerating Continuous Learning in Autonomous Systems for Video Analytics,” in *ACM/IEEE International Symposium on Computer Architecture (ISCA)*, June 2024.
5. S. Ghodrati, S. Kinzer, H. Xu, R. Mahapatra, **Y. Kim**, B. H. Ahn, D. K. Wang, L. Karthikeyan, A. Yazdanbakhsh, J. Park, N. S. Kim, H. Esmailzadeh, “Tandem Processor: Grappling with Emerging Operators in Neural Networks,” in *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, April 2024.
6. J. Hwang, M. Kim, D. Kim, S. Nam, **Y. Kim**, D. Kim, H. Sharma, J. Park, “CoVA: Exploiting Compressed-Domain Analysis to Accelerate Video Analytics,” in *USENIX Annual Technical Conference (ATC)*, July 2022.

Honors and Awards

Humantech Paper Award Gold Prize. Samsung, “Pimba: A Processing-in-Memory Acceleration for Post-Transformer Large Language Model Serving”	2025
Distinguished Artifact Award. ISCA, “DaCapo: Accelerating Continuous Learning in Autonomous Systems for Video Analytics”	2024
Scholarship for Academic Excellence. Sungkyunkwan University	Spring 2015, Fall 2016, Fall 2017
College of Software Dean’s List. Sungkyunkwan University	Fall 2016, Fall 2017

Academic Services

Artifact Evaluation Committee	ASPLOS 2024, ISCA 2025
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Experience

Engineering Intern. Software Engineering Team Jan. 2019 – Feb. 2020

Hyprsense, Burlingame, CA, USA

- Integrated a facial landmark tracker into a real-time facial motion capture SDK using ONNX Runtime.
- Developed an iOS application to map 3D character morphs onto faces using the iPhone TrueDepth camera.

Undergraduate Research Assistant. I-SURF Program Jun. 2018 – Dec. 2018

UC Irvine, Irvine, CA, USA

- Implemented GEMM computation on a FPGA device using OpenCL via Xilinx SDAccel.

Teaching

Teaching Assistant.

KAIST, Daejeon, South Korea

- CS311: Computer Organization Spring 2022, Spring 2024
- CS230: System Programming Fall 2021, Fall 2024, Fall 2025

Skills

Programming Languages: Python, C/C++, CUDA, Triton, Chisel, Verilog

Technologies and Environments: PyTorch, Ray, Docker, Git, Linux

Languages: English (Professional Working Proficiency), Korean (Native)

References

Jongse Park. Associate Professor, KAIST

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